

BASIC ALGEBRA QUIZ 1

NOTE:

Marks: 20 weightage 10 %

PROBLEMS

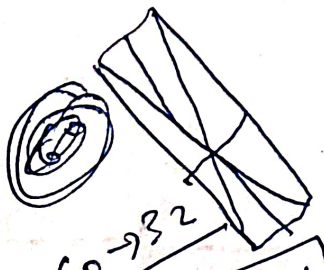
Throughout G, H, K are groups (not necessarily finite). If S is a set then $|S|$ denotes the number of elements in S .

1. Show every element in $G = \mathbb{Q}/\mathbb{Z}$ has finite order. Also show G is not cyclic [3]
2. Let G be a p -group. Show the center $Z(G) \neq \{e\}$. [5]
3. Let H be a subgroup of G . Construct a bijection from the set of left cosets of H to the set of right cosets of H . [4]
4. Let S_n be the symmetric group and let A_n be the subgroup of even permutations. Let H be a subgroup of S_n . Show that either

$$H \subseteq A_n \quad \text{or} \quad |H \cap A_n| = \frac{1}{2}|H|.$$

5. Suppose a group G only has finitely many subgroups. Show G is a finite group. [4]

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37



BASIC ALGEBRA QUIZ 2

NOTE:

Marks: 20 weightage 10 %

PROBLEMS

Throughout R denotes a commutative ring and I, J are ideals in R . Also K denotes a field. Also $|S|$ denotes the number of elements of S .

- ✓1. Show $\mathbb{Z}[i]$ is an Euclidean domain. [4]
- ✓2. Let $P_1, \dots, P_m$ be prime ideals in R . Show $S = R \setminus \bigcup_{i=1}^m P_i$ is a multiplicative closed subset of R . [3]
- ✓3. Let R be a UFD and let K be its field of fractions. Let $p(x) \in R[X]$ be a non-constant polynomial. If $p(x)$ is reducible in $K[X]$ then it is reducible in $R[X]$. [5]
- ✓4. Show $X^4 + X^3 + X^2 + X + 1$ is irreducible in $\mathbb{Z}[X]$. [4]
- ✓5. For any $n \geq 1$ Construct a PID having exactly n maximal ideals [4]

BASIC ALGEBRA QUIZ 3

NOTE:

Marks: 20 weightage 10 %

PROBLEMS

Throughout R denotes a commutative ring and M, N are R -modules. If S is a set then $|S|$ denotes the number of elements in it.

1. Let R be a P.I.D. Let $M \cong R^n$. Let N be a submodule of M . Show N is free. [5]
2. If M, N are R -submodules of T such that $T/M, T/N$ are Noetherian R -modules. Show $T/(M \cap N)$ is Noetherian. [5]
3. Let π be a prime element in $\mathbb{Z}[i]$. Show
(a) $|\mathbb{Z}[i]/(\pi)|$ is either p or p^2 for some prime p . [3]
(b) Show if $|\mathbb{Z}[i]/(\pi)| = p^2$ then $p \equiv 3 \pmod{4}$. [2]
4. Let R be a Noetherian local ring with unique maximal ideal $\mathfrak{m}$ and let M, N be finitely generated R -modules and let $f: M \rightarrow N$ be R -linear. Suppose the induced map $\bar{f}: M/\mathfrak{m}M \rightarrow N/\mathfrak{m}N$ is surjective. Show f is surjective. (Hint: Use Nakayama's Lemma) [5]

MIDSEM

NOTE

Marks: 60. Weightage 30 percent.

PROBLEMS

Throughout G, H, K are groups. A group is said to be simple if it has no normal subgroups other than $\{1\}$ and G . Throughout R denotes a non-zero commutative ring.

- ✓(1) Let H, K be two Sylow p -subgroups of a finite group G . Show H, K are conjugate. [6]
- ✓(2) Let p be a prime. Construct a non-abelian group of order p^3 having a normal subgroup isomorphic to $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$. [6]
- ✓(3) Let p, q be distinct primes. Show a group of order p^2q is NOT simple. [8]
- ✓(4) If the center of G is of index n , prove that every conjugacy class of G has at most n elements. [5]
- ✓(5) Let $|G| = p^m$ with $m \geq 2$. Let H be a subgroup of G of order p^{m-1} . Show H is normal in G . [5]
- ✓(6) Show that every group G of order 77 is cyclic. [5]
- ✓(7) Let H be a subgroup of $(\mathbb{Q}, +)$ of finite index. Show $H = \mathbb{Q}$. [8]
- ✓(8) Let $I \neq R$ be a proper ideal of R . Show there exists a maximal ideal M such that $I \subseteq M$. [7]
- ✓(9) Suppose I, J are co-maximal ideals of R . Show the ideals I^{101} and J^{311} are co-maximal. [5]
- ✓(10) Let $\mathbb{Z}_{(7)} = \{x \in \mathbb{Q} \mid x = a/b \text{ where } 7 \text{ does not divide } b\}$. Show $\mathbb{Z}_{(7)}$ has a unique maximal ideal. [5]

→ 20 marks each

BASIC ALGEBRA ENDSEM

NOTE:

Marks: 100 weightage 50%

PROBLEMS

Throughout G, H are groups. We say G is a *simple group* if it has no normal subgroups other than $\{1\}$ and G . Throughout R denotes a commutative ring and I, J are ideals in R . Furthermore M, N are R -modules.

1. Let p be a prime. Show there exists a non-abelian group of order p^4 containing a normal subgroup isomorphic to $(\mathbb{Z}/p\mathbb{Z})^3$. [7]
2. Show a group of order 15 is cyclic. [5]
3. Let H be a subgroup of a group G . Construct a bijection from the set of left cosets of H to the set of right cosets of H . [7]
4. Prove first Sylow theorem. [11]
5. Let R be a UFD. Show $R[X]$ is also a UFD. [11]
6. Prove structure theorem of finitely generated modules over a PID. [6]
7. Let $\mathbb{Q}_7 = \{x \in \mathbb{Q} \mid x = a/b \text{ where } 7 \text{ does not divide } b\}$. Show $\mathbb{Q}_7$ is a local ring. [5]
8. Give an example of ring which is NOT Noetherian. (You have to prove your assertion) [6]
9. Let $V =$ polynomials in X of degree ≤ 3 with complex coefficients. Let $T = d/dX$ be the differentiation operator. Find the Jordan form of T . [6]
10. Let R be a finite integral domain. Show R is a field. [8]
11. Let p be a prime. Let $m \geq 1$. Show there exists a field F with $|F| = p^n$ and $n > m$. [6]
12. Let $p \equiv 1 \pmod{4}$. Show p is NOT a prime in $\mathbb{Z}[i]$ [7]
13. Let R be a ring containing $\mathbb{C}$ as a subring such that $\dim_{\mathbb{C}} R = n$. Show R has at most n maximal ideals (Hint: Use Chinese remainder theorem) [5]
14. Let R be a P.I.D. Show if p is irreducible then it is a prime in R . [5]
15. Show $Y^2 + X^2 + 2X$ is irreducible in $\mathbb{Q}[X, Y]$.

$y^2 + x^2 + 2x$

root

reducible

finite integral domain

field

$\mathbb{Z}/p\mathbb{Z}$

$y^2 + x^2 + 2x$ in $\mathbb{Z}[x]$

$f = y^2 + x^2 + 2x$

$(0, 1)$

every non-zero elt has

Product Assoc

$(R/p) \rightarrow \text{field}$

$ab=0 \Rightarrow a=0 \vee b=0$

Field

$(y^2 + x^2 + 2x) \in \mathbb{Z}[x]$

$\mathbb{Z}/p\mathbb{Z}$

$\mathbb{Z}/p\mathbb{Z}$

BASIC ALGEBRA ENDDSEM

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PROBLEMS

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1. Let G be a finite group. Let P, Q be p -Sylow subgroups of G . Show P and Q are conjugate. [8] ✓
2. Let p be a prime. Show there exists a *non-abelian* group of order p^4 containing a normal subgroup of order p^3 . [7] ✓
3. Let G be a group of order p^2q where p, q are distinct primes. Also assume that order of G is NOT 12. Show G is not simple. [6] ✓
4. Let p be a prime. Find a Sylow p -subgroup of $GL_2(\mathbb{Z}/p\mathbb{Z})$. [6] ✓
5. Prove Nakayama's lemma. [6]
6. Let $I + J = R$. Show $I^{45} + J^{19} = R$. [8]
7. Let R be a Noetherian ring. Show $R[X]$ is Noetherian. [8]
8. Let R be a PID. Show R is a UFD. [6]
9. Suppose $\mathbb{Q}[X]/(f(X))$ is a local ring. Then show that $f(X) = p(x)^r$ for some irreducible polynomial $p(X)$ and $r \geq 1$. [6]
10. Suppose $M \cong N$. Show $M/IM \cong N/IN$. [7]
11. Show $\mathbb{Z}[i]$ is an Euclidean domain. [6]
12. Let $R = \mathbb{C}[X]/(X^5)$. Suppose M is a R -module such that $M = XM$. Show $M = 0$ (You cannot assume to begin with that M is finitely generated) [6]
13. Let $N \subseteq M$. Suppose N and M/N are Noetherian. Show M is Noetherian. [6]
14. Let $p \equiv 3 \pmod{4}$ be a prime. Show p remains prime in $\mathbb{Z}[i]$. [6]
15. Let p be a prime. Let $m \geq 1$. Show there exists a field F with $|F| = p^n$ and $n > m$. [8] ✓

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